

TEBEA, Romania

WMO: 152060

Lat: 46.1694N

Lon: 22.7261E

Elev: 274

StdP: 98.07

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.7	-15.0	1.1	-10.4	-12.2	1.4	-8.0	7.3	1.7	5.8	1.1	1.1	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.1	31.9	20.6	30.1	20.5	28.4	19.9	22.1	28.9	21.3	28.1	20.7	27.1	2.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.2	24.6	19.3	14.5	23.8	18.6	13.9	23.0	65.9	29.3	63.2	28.0	60.7	27.1	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.2	4.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-1.4	2.0	5.5	10.7	15.3	19.1	20.0	20.6	15.6	10.7	5.8	0.9
	DBStd	8.37	4.53	4.08	3.92	4.57	3.02	3.14	2.79	3.06	4.06	3.82	4.25	3.71
	HDD10.0	1234	354	225	147	48	2	0	0	0	3	39	136	281
	HDD18.3	3108	612	458	398	230	101	28	16	13	102	237	376	539
	CDD10.0	1394	0	1	7	67	167	273	310	327	170	61	11	0
	CDD18.3	227	0	0	0	0	7	51	68	82	18	0	0	0
	CDH23.3	2546	0	0	1	39	124	524	726	861	244	27	0	0
	CDH26.7	828	0	0	0	1	14	150	246	347	69	1	0	0
Wind	WSAvg	1.5	1.2	1.5	1.7	2.0	1.6	1.6	1.7	1.5	1.4	1.4	1.4	1.4
Precipitation	PrecAvg	691	42	35	41	56	71	83	84	51	66	59	52	54
	PrecMax	914	76	117	96	110	136	165	167	109	149	131	107	134
	PrecMin	475	14	5	10	10	32	23	21	13	4	3	15	3
	PrecStd	116	20	26	26	23	25	35	42	25	34	34	22	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	15.2	21.9	26.4	28.6	31.8	33.5	34.7	31.6	25.9	19.5	12.0
		MCWB	7.7	9.4	12.3	16.0	18.6	21.6	21.1	21.3	19.4	15.5	12.7	8.3
	2%	DB	7.6	12.6	18.4	24.3	26.4	30.1	31.5	32.5	28.5	23.6	17.0	9.3
		MCWB	5.7	8.0	10.5	14.4	17.9	21.2	20.7	20.5	18.9	15.1	11.8	7.1
	5%	DB	5.6	10.2	15.3	22.1	24.6	28.3	29.2	30.2	26.2	20.8	14.5	7.2
		MCWB	4.5	6.5	9.1	13.3	16.8	20.3	20.5	20.3	17.9	14.3	10.4	5.9
	10%	DB	3.8	7.9	12.5	19.5	22.5	26.6	27.3	28.1	23.4	17.7	12.3	5.6
		MCWB	2.8	5.4	7.5	12.2	15.9	19.7	19.6	20.0	17.1	13.0	9.6	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	10.5	12.8	17.0	20.2	23.4	22.7	23.1	21.0	17.2	13.5	9.4
		MCDB	10.4	14.5	21.1	25.0	26.6	30.2	29.2	31.1	29.3	22.7	17.3	11.5
	2%	WB	6.1	8.7	11.3	15.2	18.8	21.8	21.8	22.0	19.6	15.7	12.5	7.2
		MCDB	7.3	11.7	17.6	22.9	24.7	28.1	28.7	29.1	26.9	21.3	16.1	8.9
	5%	WB	4.7	7.3	9.7	14.1	17.8	21.0	21.2	21.2	18.6	14.7	11.4	6.1
		MCDB	5.7	9.8	14.8	21.0	23.0	27.4	27.8	28.0	24.4	20.0	14.2	7.7
	10%	WB	3.0	5.7	8.2	13.0	16.7	20.1	20.4	20.5	17.6	13.5	10.2	4.8
		MCDB	3.8	7.7	11.6	18.7	21.5	25.5	26.3	27.0	22.3	17.4	12.5	5.6
Mean Daily Temperature Range	5% DB	MDBR	7.1	9.0	11.1	12.3	11.7	12.1	12.8	13.1	12.0	11.9	9.0	6.2
		MCDBR	9.0	13.2	16.9	17.3	15.7	15.0	16.5	17.3	16.8	18.0	12.9	9.0
		MCWBR	6.5	8.4	9.3	8.3	7.9	6.8	7.0	6.6	7.3	9.8	7.9	6.7
	5% WB	MCDBR	8.4	11.9	15.3	15.5	14.4	14.1	14.2	15.0	14.6	15.7	10.7	9.1
		MCWBR	6.1	7.8	8.5	7.6	7.6	6.6	6.5	6.5	6.7	8.6	6.5	6.7
Clear-Sky Solar Irradiance	taub		0.320	0.349	0.392	0.447	0.434	0.427	0.445	0.452	0.414	0.376	0.346	0.322
	taud		2.345	2.278	2.216	2.088	2.184	2.244	2.202	2.178	2.252	2.353	2.387	2.380
	Ebn at Noon		766	811	825	809	835	842	820	796	793	772	729	721
	Edh at Noon		76	99	122	151	142	134	139	137	116	91	73	67
All-Sky Solar Radiation	RadAvg		1.24	2.04	3.17	4.33	5.20	5.78	5.67	5.21	3.71	2.57	1.49	1.00
	RadStd		0.17	0.26	0.46	0.46	0.49	0.51	0.49	0.51	0.41	0.30	0.20	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (65 neighbors)	+0.70	N/A	N/A	+0.97	N/A	N/A	-125	-189	+121	+60

Nomenclature: See separate page